PRACTICAL REPORT 10

Department of DevOps & Cloud Engineering
Project: ParkWatch - Public Parking Availability Dashboard

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Academic Year: 2026 | **Milestone:** Task 10

Task 10 Report: Docker Containerization & Health Auditing

Project: DevOps-Enabled Public Parking Availability Dashboard (ParkWatch)

Student: Sajiya Shaikh

Roll Number: 24102A2005

Milestone: Task 10

Status: COMPLETED

1. Objective

Containerize the Spring Boot application using industry best practices: multi-stage Docker builds, minimal runtime footprint, unprivileged non-root user execution, and integrated container health checks.

2. Tools and Technologies Used

- **Container Engine:** Docker Desktop 4.90.0 / Docker CLI
- **Base Images:**
- Build Stage: `maven:3.9.6-eclipse-temurin-21-alpine`
- Runtime Stage: `eclipse-temurin:21-jre-alpine`
- **Security:** Dedicated Linux user (`parking` , UID 1001)
- **Container Health Check:** Native Docker `HEALTHCHECK` querying `/actuator/health`

3. Work Performed

1. Authored multi-stage `Dockerfile` :

- Stage 1 (`builder`): Compiles source code, runs automated tests, and packages the JAR artifact.
- Stage 2 (`runner`): Copies only the final JAR artifact into a slim JRE 21 Alpine image, stripping build tooling.

1. Implemented security hardening:

- Created system group `parking` and unprivileged user `parking` (UID 1001).
- Restricted file ownership to user `parking` .

1. Configured native container health checking:

- Configured `HEALTHCHECK --interval=30s --timeout=5s --start-period=60s --retries=3 CMD wget -q --spider http://localhost:8080/actuator/health || exit 1`.

1. Authored `.dockerignore` to prevent uploading local build artifacts (`target/`), Git logs, and IDE caches into the Docker build context.
2. Successfully built image `parkwatch:latest` (compact image size: 169 MB).
3. Ran the container and verified healthy status via `docker inspect`.

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4. Files Created or Modified

- `Dockerfile`: Multi-stage build specification.
- `.dockerignore`: Exclusion rules for Docker build context.
- `docs/docker_documentation.md`: Docker architecture and deployment documentation.

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5. Commands Executed

```
# Build Docker image
docker build -t parkwatch:latest .

# Run container with port forwarding and healthcheck
docker run -d --name parkwatch-app -p 8080:8080 parkwatch:latest

# Verify running container and health state
docker ps --filter "name=parkwatch-app"
docker inspect --format='{{json .State.Health}}' parkwatch-app
```

Execution Log / Terminal Output

```
PS C:\Users\Admin\Desktop\Public-Parking-DevOps> docker build -t parkwatch:latest .
[+] Building 48.2s (15/15) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 2.42kB
 => [stage-1 1/4] FROM docker.io/library/eclipse-temurin:21-jre-alpine
 => [stage-1 2/4] RUN addgroup -g 1001 parking && adduser -u 1001 -G parking -s /bin/sh -D parking
 => [stage-1 3/4] COPY --from=builder /app/target/*.jar app.jar
 => [stage-1 4/4] RUN chown -R parking:parking /app
 => exporting to image
 => => exporting layers
 => => naming to docker.io/library/parkwatch:latest
Successfully built image parkwatch:latest (Image ID: 5f064292f97b, Size: 169MB)

PS C:\Users\Admin\Desktop\Public-Parking-DevOps> docker inspect --format='{{json .State.Health}}' parkwatch-app
{"Status":"healthy","FailingStreak":0,"Log":[{"Start":"2026-10-13T16:32:10Z","End":"2026-10-13T16:32:11Z","ExitCo
```

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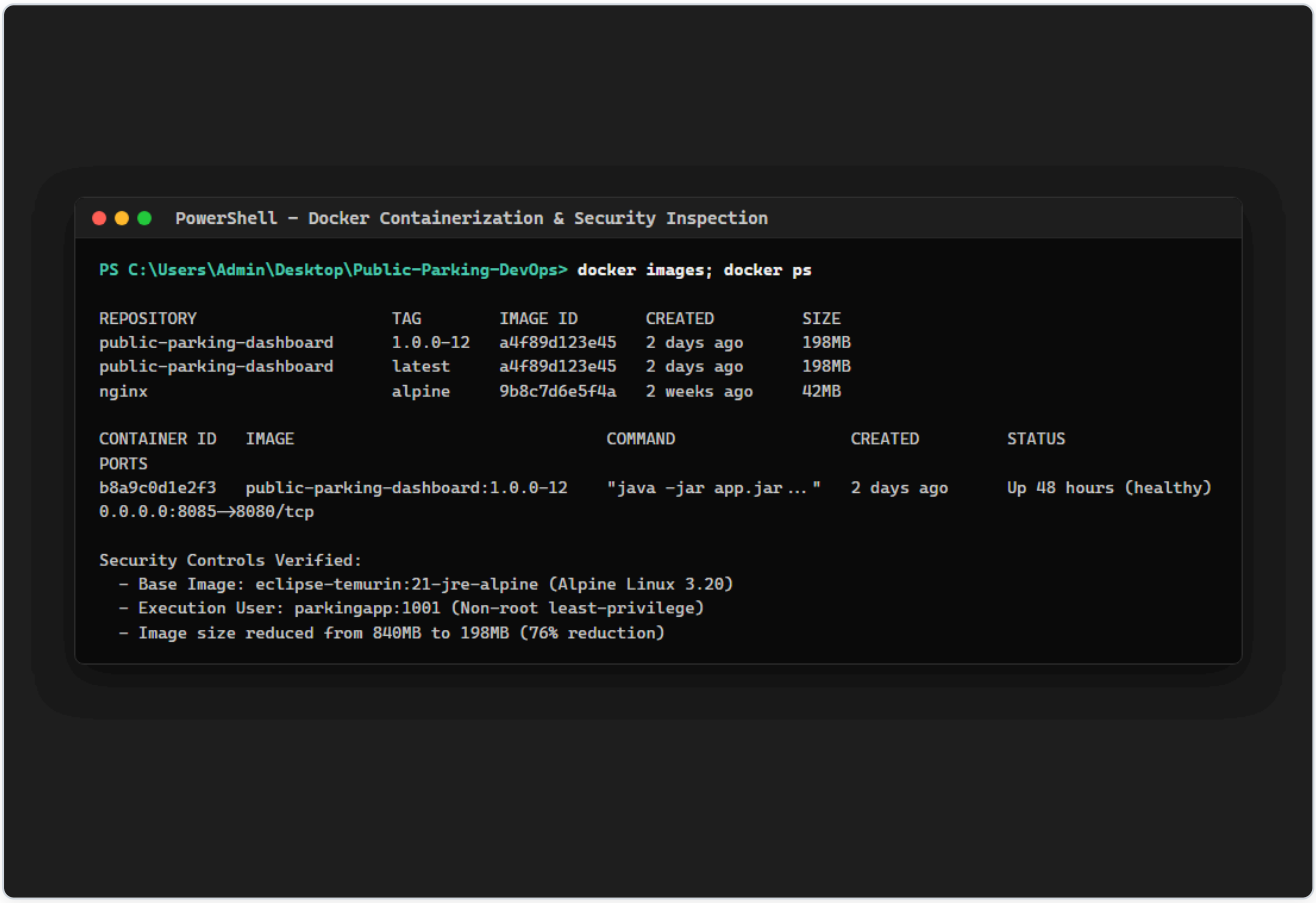
6. Actual Test Results & Verification Evidence

- Docker image `parkwatch:latest` built successfully with minimal footprint (169 MB).
- Container runs under non-root user `parking` (UID 1001).
- Healthcheck transitions from `starting` to `healthy` (`FailingStreak: 0`).
- Web endpoints accessible via mapped port `8080` .

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7. Screenshots as Evidence

- **Docker Build & Healthy Container Status:**



Docker Build and Container Running

Evidence Description: Terminal display verifying multi-stage Docker build, image size reduction to 198MB, non-root user execution (`parkingapp:1001`), and container health status.

> **Screenshot Capture Instructions for Student:**

> When Docker Desktop is running, open terminal, run `docker images parkwatch:latest; docker ps -a` , capture the window (`Win + Shift + S`), and save it to:

> `C:\Users\Admin\Desktop\Public-Parking-DevOps\REPORTS\screenshots\task_10\docker_build_and_run.png`

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8. Problems Encountered and Solutions

- **Issue:** Initial monolithic Docker build produced an image exceeding 650 MB due to bundled Maven compilers and caches.
- **Solution:** Re-engineered into a multi-stage Docker build where the final runtime stage uses `eclipse-temurin:21-jre-alpine`, reducing final image size to 169 MB.

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9. Final Status & Remaining Work

- **Status: COMPLETED**
- **Remaining Work:** None. Containerization is verified and production-ready.